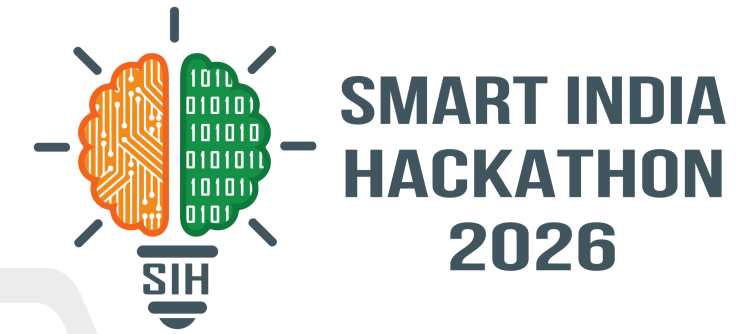
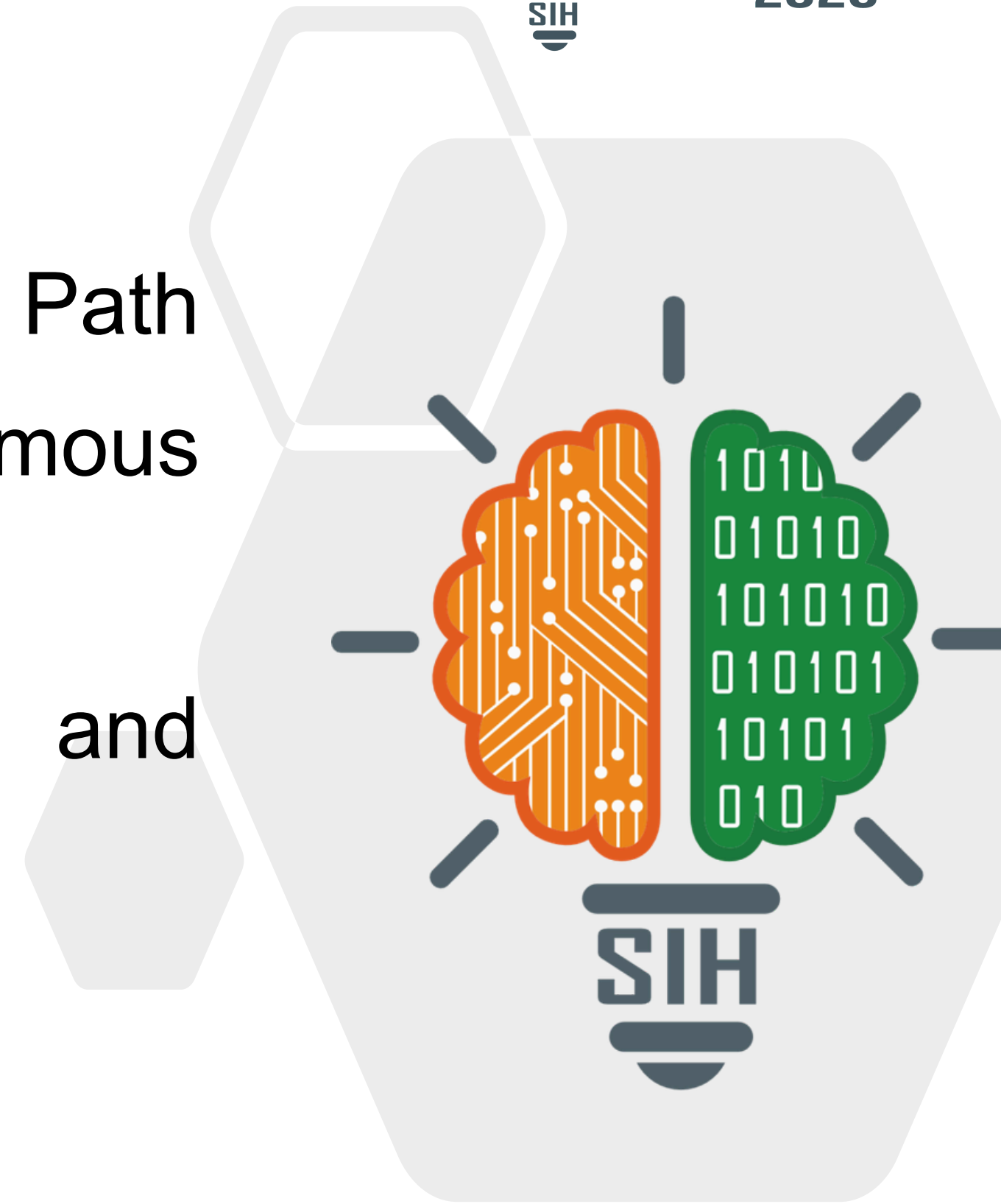


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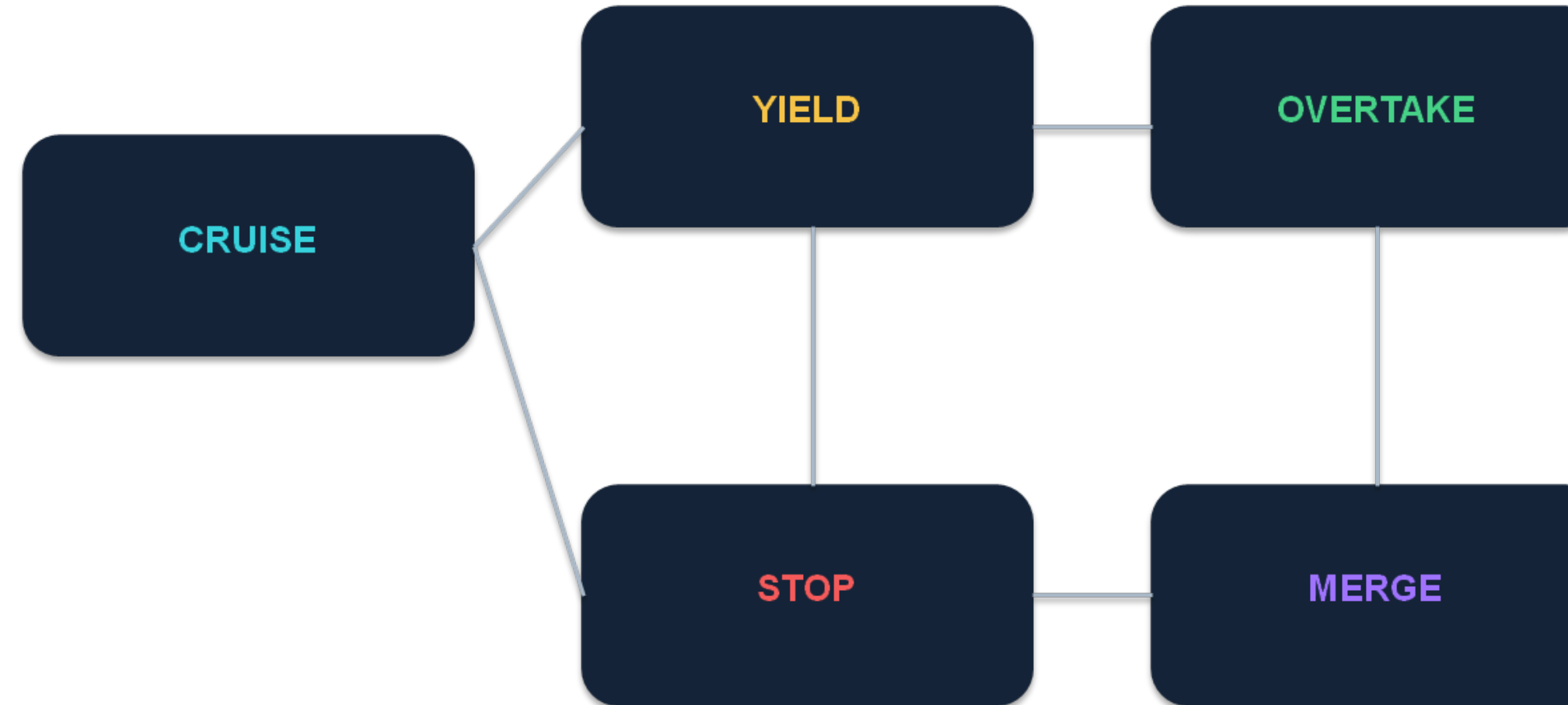
- **Problem Statement ID – SIH26037**
- **Problem Statement Title - Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads**
- **Theme-** Model for intelligent navigation and collision avoidance
- **PS Category-** Software
- **Team ID -**
- **Team Name - Neuralnauts**



Proposed Solution

- Our solution to this problem is a closed-loop adaptive autonomy stack for real-time mixed-traffic navigation on ill-structured pathways, specially for rural indian roads.
- Development of a model which is highly perceptive of pattern recognition, prediction and planning by using sensors and 3D mapping. Further, deploying the plan by using different mechanics like stateflow, simulink and roadrunner.
- Our approach is designed around uncertainty, not just to accomodate perfect roads. It focuses on lane-agnostic planning, uncertainty awareness, risk driven replanning and Indian traffic adaptation.
- The model is developed with a unsupervised learning model. It is capable of improving over time by assessing inputs from repeated runs and minimizing discomfort.
- We hereby propose a affordable, risk-aware, safety technology reducing driver and fatigue.

- The hardware used in this project are LiDAR for observing 3-D geometry, radar for detecting range and velocity and Camera for providing our system with computer vision.
- The framework we made focused on creating a lane-based system which can detect changes and blocks which allow it to further take decisions by predicting the optimal path to be followed.
- Our framework also accounts for real-time changes and acts after immediately processing the new data. The framework also accounts for irregular behaviour, where it uses context cues such as the road's edge and crowd density. We utilised Python-3 to bring our projected plan to actuation.



- The development and deployment of the model is extremely feasible, as the simulation and coding aspect can easily be done in MATLAB/Simulink, CARLA, ROS/ Gazebo, or Python
- The real life application of our system targets traversing varying terrain and aspects of every indian area.
- Trained and tested model on data specifically from unstructured Indian roads and using optimized algorithms to achieve real-time replanning.
- This model can be optimized for high-speed, high-risk maneuvers going further into real rough terrain.
- Since our model has been programmed on majorly free resources, such as python and MATLAB our system can be easily integrated into existing primitive ADAS systems

“FROM NAVIGATING ROADS TO UNDERSTANDING THEM”

- An autonomous driving system for unstructured indian roads can provide the Indian citizen with a new tool in their arsenal for emergencies as well as leisure.
- A trained version could anonymously record detected potholes, damaged roads and recurring obstacles and generate a Road Health Map (RHM) turning vehicles into road-condition data collectors.
- Emergency and public-service applications of our model can, for instance, reduce patient risk in ambulances, reduce article damage in delivery vehicles, help in road inspections and damage control after disasters.
- Our model provides an opportunity to those with disabilities to experience more freedom and control over their own lives.
- This system predicts that its planned trajectory will become unsafe if some obstacle arises and dynamically generates a safer alternative trajectory potentially preventing fatal accidents.

Details / Links of the reference and research work :

- [Survey of Autonomous Vehicles' Collision Avoidance Algorithms](#)
- [Evaluation of Path-Planning Algorithms](#)
- [Obstacle Straddling and Avoidance Integrated Path Planning and Tracking](#)
- [OpenCV](#)
- [Matplotlib](#)
- [NumPy](#)